# Unraveling the mysteries of gene transcription activation with bioinformatics

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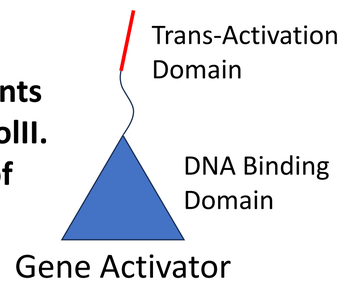
## We don't *really* understand how transcription activation domains (TAD's) work.

**"In freshman biology classes, we teach the Central Dogma: DNA → RNA → protein. These lessons are accompanied by cartoon drawings of a transcription factor binding DNA and recruiting components of the transcriptional machinery, such as Mediator and PolII. However, these cartoons imply a deeper understanding of this process than, in reality, exists."**

- Strader L. *et al. Mol Cell* 2023; 83(3): 320-323.

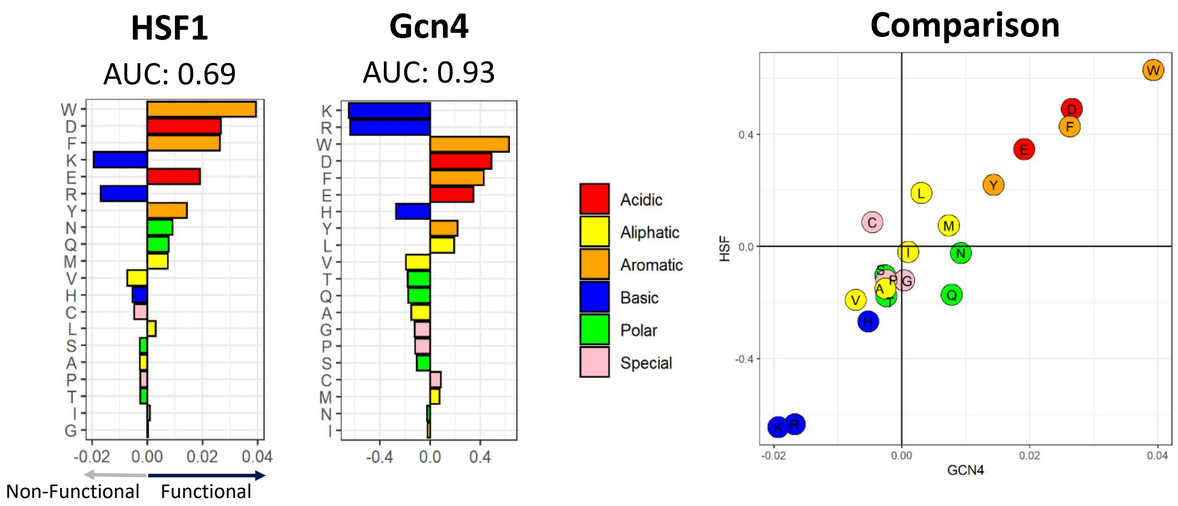



- The DNA Binding Domain has a very specific sequence and structure.
- The Trans-Activation Domain has **no specific sequence or structure.**

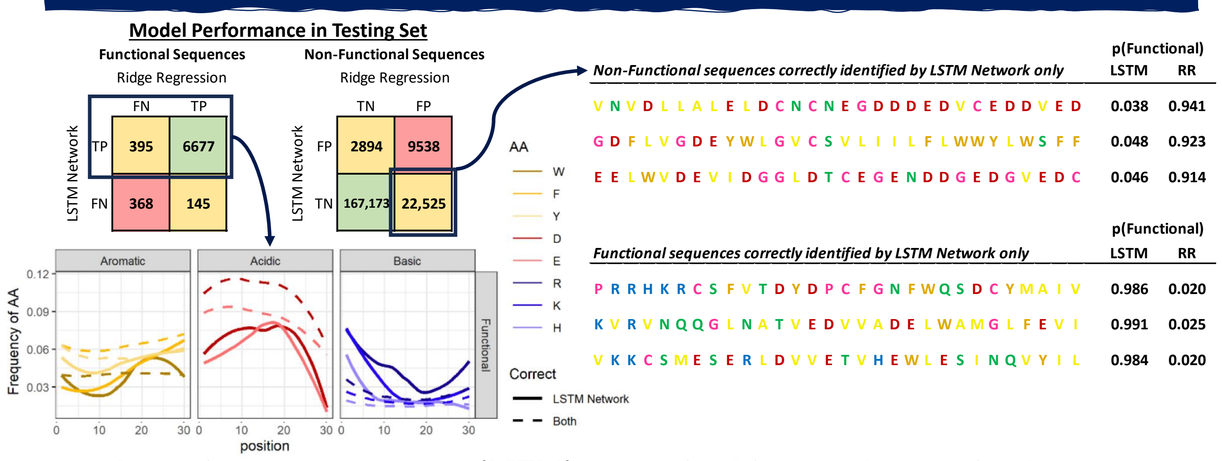
***What defines a functional TAD? And how does it really work?***

## Random amino acid sequences have a >1% chance of functioning as a transcription activation domain.

- Two independent studies profiled random peptide sequences as TAD's.
  - Ravarani *et al.* (2018) – 1.1% of 67,000 sequences were functional in HSF1.
  - Erijman *et al.* (2020) – 3.7% of 1 million sequences were functional in Gcn4.
- **Acidic** and **aromatic** residues are beneficial for function in both studies, while **basic** residues are detrimental.
  - Ridge regression: Functionality (1/0) ~ # of each AA

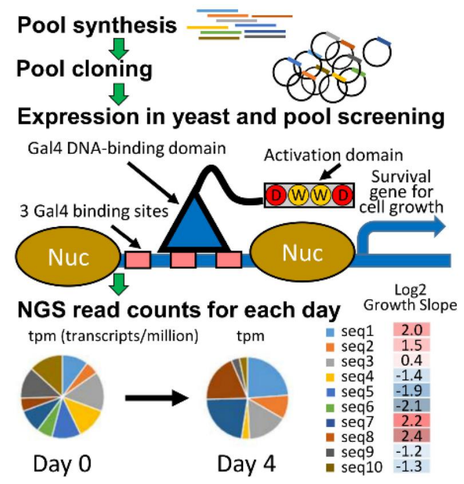



## Machine learning identifies patterns that simple models don't.



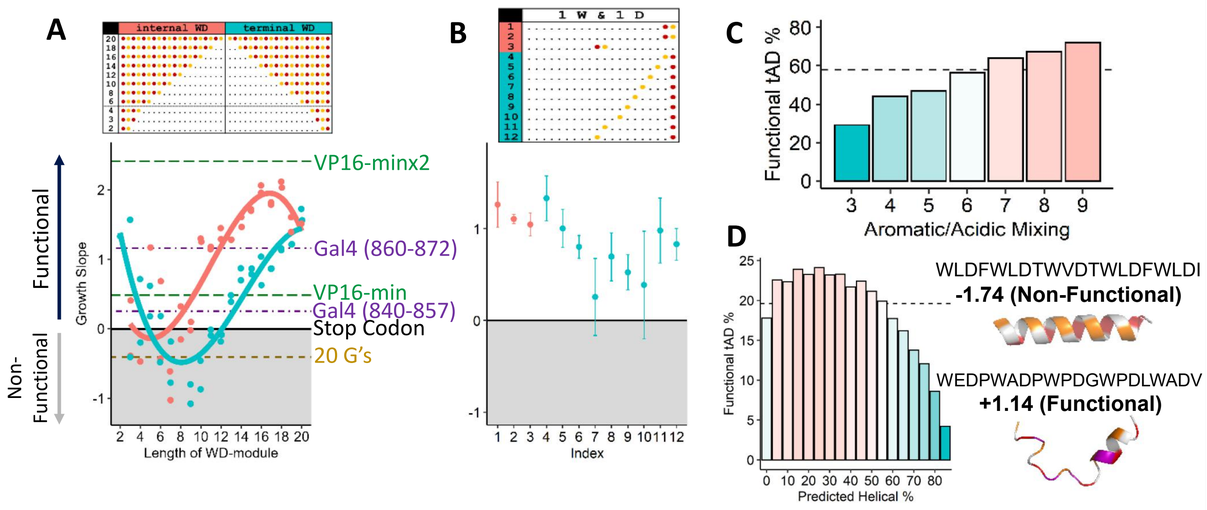

- Long short-term memory (LSTM) network with attention mechanism
  - **LSTM network**: Neural network that analyzes complex sequence relationships
  - **Attention mechanism**: Analyzes contribution of each AA to functionality
- For the Gcn4 library, LSTM network improves AUC from 0.93 to 0.98
- **Rules for TAD functionality based on random sequence libraries**
  - ✓ Both aromatics & acidic amino acids are required
  - ✓ Basic amino acids are acceptable (but not necessary) at 5' end
  - ✓ Aromatics preferred at 3' end
  - ✓ Acidic amino acids in between

## Designed sequence libraries help us understand the patterns.




- 11,500 designed TAD sequences synthesized by GenScript to test TAD functionality "rules"
  - Number and position of aromatic/acidic AA's
  - Balance of aromatic/acidic resiidues
  - Mixing of aromatic/acidic residues
  - STOP codon sequences as negative control
- 108,000 semi-random TAD sequences ("WD5")
  - WxDxWxDxWxDxWxDxWxDx (**x** = random)
  - Test functionality with/without α-helix

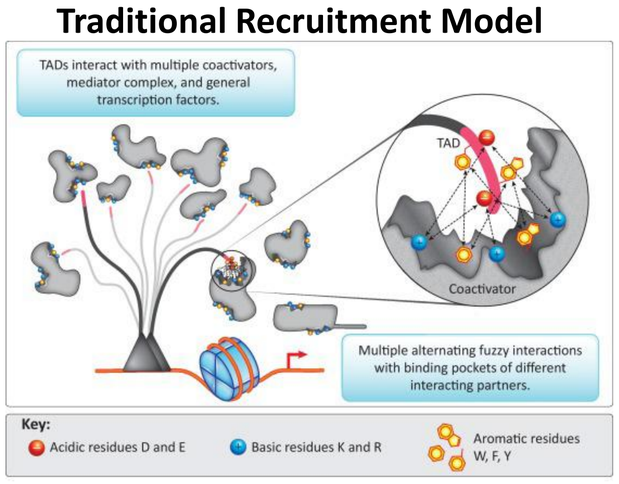
## *One aromatic and one acidic residue* are sufficient for TAD functionality.



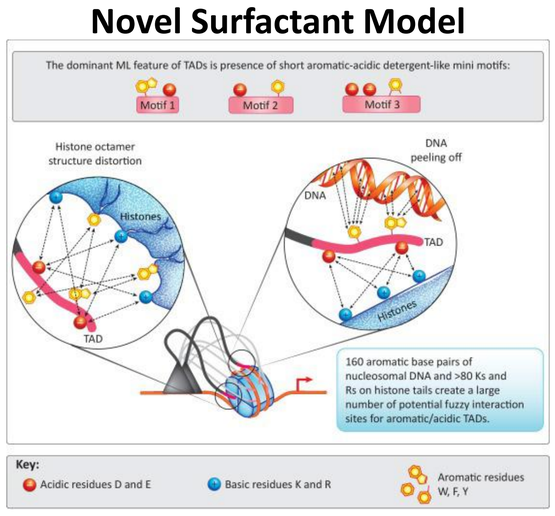

- **(A-B)** A single **tryptophan (W)** and **aspartic acid (D)** residue are sufficient when included among 18 glycines.
- **(C)** For TAD's with 6 W's and 6 D's, **highly mixed TAD's** are more likely to function.
- **(D)** In the WD5 (semi-random) library, **low helicity** is associated with greater TAD function. Prolines support functionality by acting as "helix-breakers".

## Interpretation (The hard part)

### Traditional Recruitment Model




### Novel Surfactant Model




- **Traditional recruitment model** requires specificity, prefers helicity.
- **Surfactant model** allows for non-specific interactions.
- **Liquid-Liquid Phase Separation (LLPS)** supported by **"stickers and spacers"**.
  - Hydrophobic stickers are exposed by hydrophilic spacers.

***What do you think?***

## References

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